

Appendix A. Stochastic conformance checking with guarantees – Proofs

This document contains formal proofs that justify the proposed stochastic conformance checking approach.

Proposition 1 (Well-Formedness). *The intersection of two well-formed FLMCs is a well-formed FLMC.*

Proof. By construction, the intersection model contains only reachable states, as these correspond to pairs of reachable states of the original models. Dead states are removed according to the definition of intersection. The resulting model is also split by construction, since transitions on different symbols lead to different state pairs, as they lead to different states in the original models. Finally, the empty sequence is not accepted by the intersection model, since it is not accepted by either of the original models. $\square$

Property 2 (Representation). *A stochastic-aware conformance measure should depend on the stochastic languages of logs and models and not on their representations.*

Proof. Different FLMC representations of the same stochastic language may contain several equivalent states that correspond to the same stochastic language generated when starting in these states. Such states can be merged without changing the represented stochastic language.

Let $s_1, \dots, s_k$ be equivalent states that are merged into a single state s . The stationary probability of s in the merged FLMC equals the sum of their stationary probabilities:

$$\pi(s) = \sum_{i=1}^k \pi(s_i).$$

Hence, when either of the models with states $s_1, \dots, s_k$ or single state s intersects with another model with a corresponding state r , for the conformance measures the value of

$$\pi(r) - \sum \pi(r, s_i),$$

equals

$$\pi(r) - \pi(r, s).$$

Therefore, merging equivalent states does not change the precision and recall measures. Conversely, splitting a state into equivalent states does not change it either.

□